

Advanced Lesson 13 - Exercises:

Trace and Held Evaluation

Mikey Ferguson

MATH 7013 - Advanced Mathematical Modeling

1: Tracing differentiation and integration

```
Clear[x];  
Trace[ $\partial_x x^2$ ]  
Trace[ $\partial_x (1 + x)^2$ ]  
Trace[ $\partial_x (x + 1)^2$ ]  
Trace[ $\int (x + x^2 + e^x) dx$ ]  
Trace[ $\partial_x \int (1 + x)^2 dx$ ]
```

2: Tracing immediate against delayed replacement rules

```
Clear[a, b];  
Trace[Sin[a] /. a -> b]  
  
Clear[a, b];  
b = 5;  
Trace[Sin[a] /. a -> b]  
  
Clear[a, b];  
a = 2;  
b = 5;  
Trace[Sin[a] /. a -> b]
```

```

Clear[a, b];
Trace[Sin[a] /. a -> b]

Clear[a, b];
b = 5;
Trace[Sin[a] /. a -> b]

Clear[a, b];
a = 2;
b = 5;
Trace[Sin[a] /. a -> b]

```

3: Displaying arguments unevaluated and evaluated, side by side

```

ClearAll[f, a];
SetAttributes[f, HoldAll];
f[args___] := List @@ Function[u, {HoldForm[u], u}, HoldAll] /@ Hold[args]

Clear[a];
f[N[ $\pi$ ], 1 + 2, a  $a^2$ , a (1 + a)]

TableForm[f[N[ $\pi$ ], 1 + 2, a  $a^2$ , a (1 + a)]]

```